SUTEX BANK COLLEGE OF COMPUTER APPLICATION & SCIENCE

PROJECT REPORT

FLIGHT PRICE PREDICTION USING MACHINE LEARNING

Submitted in partial fulfillment of the practical work for

303 – 05 Fundamentals of Machine Learning (PR)

Course / Semester	SYBCA-AI, Semester III
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Academic Year	2026–2027

Sutex Bank College of Computer Application & Science

Surat, Gujarat

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his/her Practical work in Paper No. **303-05 Fundamentals of Machine Learning (PR)** *for the*

Semester **III** *during the academic year* **2026-27**

Supervisor :

Examiner :

Date : 07/10/26

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DATASET DESCRIPTION

1. Dataset Overview:-

The Flight Price Prediction dataset contains information about flight journeys and the factors that may affect ticket prices. The project uses journey-related attributes such as the number of stops, date, month, arrival time, departure time and journey duration. The target variable is Price, which represents the flight ticket price. The dataset is analyzed and preprocessed before applying regression algorithms to estimate flight prices.

2. First 5 Rows:- df.head()

The first five records can be displayed using df.head(). The actual table is available in the dataset output.

3. Column Names:- df.columns

The column names are inspected using df.columns. The supplied correlation matrix includes Total_Stops, Price, Date, Month, Year, Arrival_minute, Dept_min and duration_minute.

4. Dataset Information:- df.info()

The dataset structure and data types are checked using df.info().

5. Statistical Description:- df.describe()

Numerical summaries such as count, mean, standard deviation, minimum, maximum and quartiles are generated using df.describe().

6. Dataset Shape:- df.shape

The exact row and column count is obtained directly from df.shape. A numeric count is not stated here because it is not visible in the supplied project outputs.

7. Missing Value Analysis:- df.isnull().sum()

Missing values are checked for every column using df.isnull().sum().

8. Duplicate Value Analysis:- df.duplicated().sum()

Duplicate records are checked using df.duplicated().sum().

9. Unique Value Analysis:- df.nunique()

The number of unique values in each column is checked using df.nunique().

10. Numerical and Categorical Columns

Numerical columns are identified using select_dtypes(include=['int64','float']). Categorical columns are identified using select_dtypes(include=['object','category']).

11. Target Variable:- Price 12. Independent Variables:-

- Total_Stops
- Arrival_minute
- Date
- Month
- Dept_min
- duration_minute

13. Dataset Summary:-

The project uses the flight price dataset loaded from dataset.csv. The target variable is Price. In the submitted feature-selection code, the independent variables are Total_Stops, Arrival_minute, Date, Month, Dept_min and duration_minute. The Year column also appears in the supplied correlation-matrix visualization, but it is not included in the final X selection shown in the submitted report code.

DATASET DESCRIPTION (PROJECT DETAILS)

Purpose of the Dataset

The dataset is used to study how journey details relate to flight ticket prices. Each record represents a flight option or journey observation, while **Price** is the value the project aims to estimate. The dataset is loaded from `dataset.csv` into a Pandas DataFrame so it can be inspected, cleaned and prepared for model training.

Role of the Selected Columns

Total_Stops describes the number of stops in the journey. **Arrival_minute** and **Dept_min** represent arrival and departure time values in minute-based form. **Date** and **Month** capture calendar information, while **duration_minute** represents journey duration in minutes. These features are used as inputs (X), and **Price** is used as the target output (y). The report's selected feature list does not include the Year column.

Why Dataset Inspection Is Important

Before training, the project checks the number of rows and columns, column names, data types, missing values, duplicate records, unique values and statistical summaries. These checks help identify data-quality issues and confirm that the selected columns are available for analysis. The actual row and column count should be read from `df.shape` rather than assumed.

Preprocessing and Use in Prediction

The project applies the Interquartile Range (IQR) method to filter unusual values in `duration_minute`. It then separates the features and target and splits the data into 80% training and 20% testing subsets using `random_state=42`. The training subset is used to fit regression models; the test subset is reserved for evaluating how well their price estimates match the observed prices.

Project Relevance

A flight-price estimate can help a traveller compare possible journey options and understand how time, duration and stops may be associated with ticket cost. This model provides an estimate based on the patterns in the supplied dataset; it should not be treated as a guaranteed live fare because prices can also change with airline, route, booking time, demand, season and other factors that may not be included in the selected features.

Dataset inspection commands

```
df.head()    df.columns    df.info()    df.describe() df.shape
df.isnull().sum()    df.duplicated().sum()    df.nunique()
```

DATA PREPROCESSING

Data preprocessing was performed to inspect and prepare the Flight Price Dataset for Machine Learning. The submitted code checks missing values, duplicate records, unique values, data types and statistical summaries. It also removes outliers from the duration_minute column using the Interquartile Range (IQR) method.

1. Data Inspection

The project checks missing values, duplicate rows, unique values, dataset shape, column names, first and last records, data types and descriptive statistics. These checks help understand the quality and structure of the dataset.

2. Outlier Handling

Outliers in duration_minute are identified using the Interquartile Range (IQR) method. Q1 and Q3 are calculated, and values outside the lower and upper bounds are removed.

```
Q1 = df['duration_minute'].quantile(0.25)
Q3 = df['duration_minute'].quantile(0.75)
IQR = Q3 - Q1

lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

df = df[(df['duration_minute'] >= lower_bound) &
(df['duration_minute'] <= upper_bound)]
```

3. Feature Selection

Price is selected as the dependent variable (y). Total_Stops, Arrival_minute, Date, Month, Dept_min and duration_minute are selected as independent variables (X).

4. Train-Test Split

The dataset is divided into training and testing sets using an 80:20 ratio. The random_state is set to 42 so that the split can be reproduced.

```
X = df[['Total_Stops', 'Arrival_minute', 'Date',
'Month',        'Dept_min', 'duration_minute']] y =
df['Price']

from sklearn.model_selection import
train_test_split X_train, X_test, y_train, y_test
= train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) is used to understand the distribution of data and the relationships between numerical attributes. In this project, a histogram, box plot and correlation matrix were generated to study duration_minute and the numerical features.

1. Distribution of Duration Minute (Histogram)

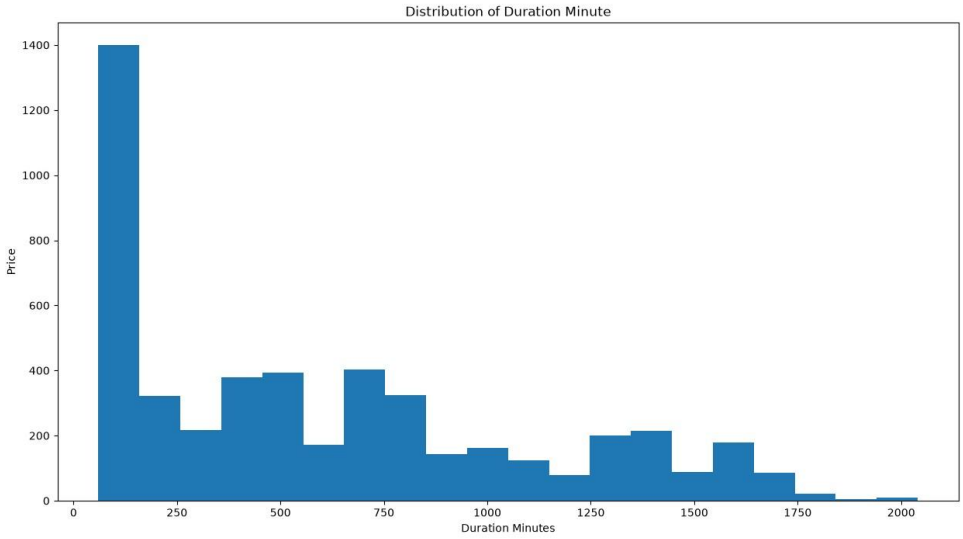


Figure 1: Histogram showing the distribution of duration_minute.

The histogram displays how the duration_minute values are distributed across the dataset. **2.**

Box Plot of Duration Time

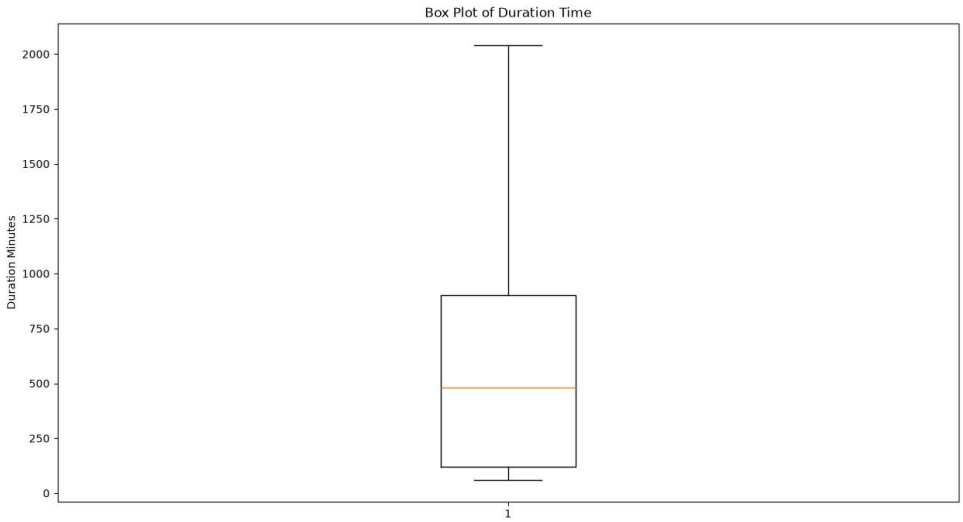


Figure 2: Box plot of duration_minute after IQR-based outlier handling.

The box plot summarizes the median, quartiles and spread of duration_minute values after the IQR filter.

3. Correlation Matrix

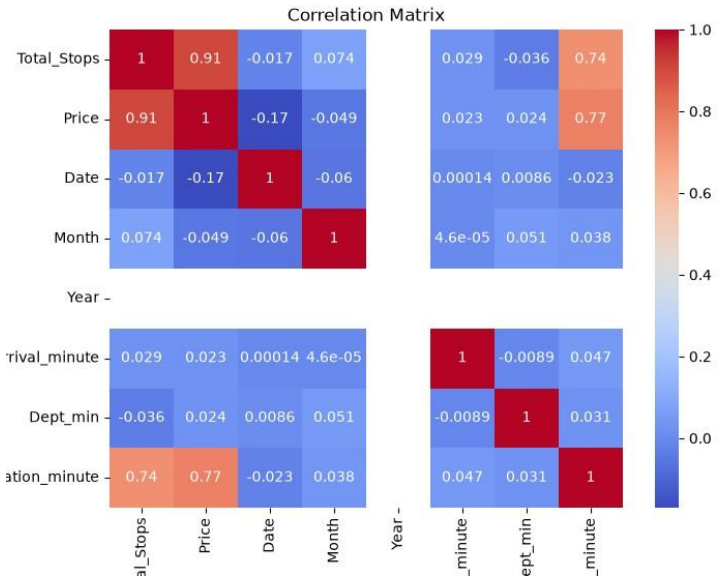


Figure 3: Correlation matrix of numerical columns.

The heatmap presents correlations among numerical attributes. The displayed values help identify the strength and direction of linear relationships.

4. Automated EDA

Sweetviz is imported and used to generate an automated exploratory data analysis report in HTML format. The report provides an additional overview of the dataset.

MODEL BUILDING (CODE)

1. Linear Regression

Linear Regression is a supervised Machine Learning algorithm used to predict continuous numerical values. In this project, it is used to estimate flight ticket prices based on the selected journey-related features. The model is trained using the training dataset and then used to predict prices for the testing dataset.

Python Code:-

```
from sklearn.linear_model import LinearRegression

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
```

2. Cross Validation

Five-fold cross validation is used to evaluate the stability of the Linear Regression model. The scoring parameter is set to R^2, and the mean of the five scores is calculated.

```
from sklearn.model_selection import
cross_val_score from sklearn.linear_model import
LinearRegression

model = LinearRegression()
cv_scores = cross_val_score(
model, X, y, cv=5, scoring='r2' )

print("Cross Validation R² Scores:")
print(cv_scores)
print("Mean R²:", cv_scores.mean())
```

3. Ridge Regression

Ridge Regression is a regularized form of Linear Regression. It applies an L2 penalty to reduce the effect of large coefficients. In this project, Ridge Regression with alpha=1.0 is trained and compared with Linear Regression. `from sklearn.linear_model import Ridge`

```
model = Ridge(alpha=1.0)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

Both models are evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE) and R^2 score.

MODEL DEVELOPMENT — CODE EXPLAINED LINE BY LINE

Step 1: Select Features and Target

The independent variables are stored in X. The target variable Price is stored in y.

```
X = df[['Total_Stops', 'Arrival_minute', 'Date', 'Month',
'Dept_min', 'duration_minute']] y = df['Price']
```

- **X** contains the journey details used to make a prediction.
- **y** contains the actual ticket prices the model learns to predict.

Step 2: Split the Dataset

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)
```

- **train_test_split** divides the data into training and testing sets.
- **test_size=0.2** keeps 20% for testing and 80% for training.
- **random_state=42** makes the split repeatable.

Step 3: Train Linear Regression

```
from sklearn.linear_model import LinearRegression
model = LinearRegression() model.fit(X_train,
y_train) y_pred = model.predict(X_test)
```

- Import **LinearRegression** from Scikit-learn.
- Create the model object using **LinearRegression()**.
- **fit()** learns the relationship between the selected features and prices from training data.
- **predict()** estimates prices for the unseen test features.

Step 4: Train Ridge Regression

```
from sklearn.linear_model import Ridge
ridge_model = Ridge(alpha=1.0)
ridge_model.fit(X_train, y_train) ridge_pred
= ridge_model.predict(X_test)
```

- Ridge is a regression model that adds L2 regularization.
- **alpha=1.0** sets the regularization strength.
- The model is fitted on the training data and then predicts values for X_test.

Step 5: Evaluate the Predictions

```
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
mae = mean_absolute_error(y_test, y_pred) mse = mean_squared_error(y_test,
y_pred) rmse = mse ** 0.5 r2 = r2_score(y_test, y_pred) print('MAE:', mae)
print('MSE:', mse) print('RMSE:', rmse) print('R2:', r2)
```

- **MAE** gives the average absolute prediction error.
- **MSE** averages the squared errors, so large errors have more influence.
- **RMSE** is the square root of MSE and is expressed in the target's units.
- **R^2** indicates how much variation in prices is explained by the model; closer to 1 is generally better.

Step 6: Five-Fold Cross Validation

```
from sklearn.model_selection import cross_val_score cv_scores =
cross_val_score(LinearRegression(), X, y, cv=5, scoring='r2')
print(cv_scores) print('Mean R2:', cv_scores.mean())
```

The data is evaluated across five folds. Each fold acts as a validation set once, and the mean R^2 summarizes the scores. In the supplied report output, Linear Regression achieved $R^2 \approx 0.8856$ on the test set and mean cross-validation $R^2 \approx 0.8848$. The reported Linear Regression result is slightly better than the Ridge Regression result.

MODEL EVALUATION

Model evaluation is used to measure how closely the predicted flight prices match the actual prices. The project uses MAE, MSE, RMSE and R^2 score.

1. Mean Absolute Error (MAE)

MAE measures the average absolute difference between actual and predicted prices. A lower MAE indicates smaller average prediction errors.

2. Mean Squared Error (MSE)

MSE calculates the average of squared prediction errors. Larger errors receive a higher penalty.

3. Root Mean Squared Error (RMSE)

RMSE is the square root of MSE. It expresses the prediction error in the same unit as the target variable, Price.

4. R^2 Score

R^2 score indicates how much of the variation in flight prices is explained by the model. A value closer to 1 indicates a stronger fit.

Python Code:-

```
from sklearn.metrics import
(     mean_absolute_error,
mean_squared_error,
r2_score )

mae = mean_absolute_error(y_test,
y_pred) mse =
mean_squared_error(y_test, y_pred) rmse
= mse ** 0.5 r2 = r2_score(y_test,
y_pred)

print("MAE:", mae)
print("MSE:", mse)
print("RMSE:", rmse)
print("R2:", r2)
```

RESULTS

The project evaluates Linear Regression and Ridge Regression using the test dataset. The following values are transcribed from the supplied output screenshots.

1. Linear Regression Results

Metric	Result
MAE	821.592412088128
MSE	1030322.199495002
RMSE	1015.0478803953054
R^2 Score	0.885621082572326

The Linear Regression model achieved an R^2 score of approximately 88.56%. Its five-fold cross-validation mean R^2 score was 0.8847781184215611 (approximately 88.48%).

2. Ridge Regression Results

Metric	Result
MAE	821.6950885925147
MSE	1030465.3234647289
RMSE	1015.1183790399664
R^2 Score	0.8856051939845393

MODEL OUTPUT

The following screenshots are included as supporting evidence of the model evaluation and cross-validation results.

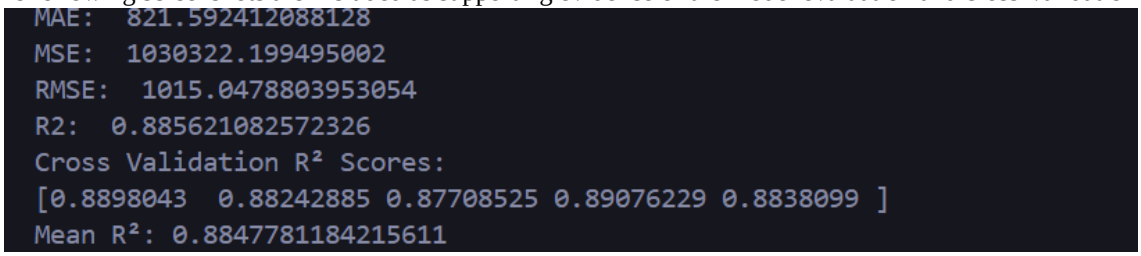


Figure 4: Linear Regression evaluation and cross-validation output.



Figure 5: Ridge Regression evaluation output.

CONCLUSION

The Flight Price Prediction project was implemented using Machine Learning regression techniques. The dataset was inspected and analyzed, and outliers in duration_minute were handled using the IQR method. Exploratory Data Analysis was performed using a histogram, box plot, correlation matrix and Sweetviz report. Linear Regression and Ridge Regression were trained to predict flight ticket prices using journey-related features. Based on the supplied evaluation outputs, Linear Regression achieved an R^2 score of approximately 88.56%, with a five-fold cross-validation mean R^2 score of approximately 88.48%. The results show that the selected features provide useful information for estimating flight prices.

FUTURE SCOPE

In the future, the project can be improved by including additional factors such as airline, source city, destination city, departure time, arrival time, booking lead time, season and holidays, where these columns are available. More advanced regression algorithms can also be tested and compared. The model can be integrated into a simple web application so that users can enter journey details and receive an estimated ticket price. Regular updates with recent flight data may help the system adapt to changing ticket-price patterns.